TD2

Ex4

On considère le problème de Laplace suivant :

$$\begin{cases} -\Delta u(x,y) = 0, & (x,y) \in \Omega :=]0,1[\times]0,2\pi[\\ u(x,0) = u(x,2\pi) = 0, & 0 < x < 1 \\ u(0,y) = 0, \ \dfrac{\partial u}{\partial x}(0,y) = -ne^{-\sqrt{n}} \sin(ny), & 0 < y < 2\pi \ (n \geq 1 \text{ fixé}) \end{cases}$$

a) Vérifier que $u(x,y) := -e^{-\sqrt{n}} sh(nx) \sin(ny)$ est solution

On a :
$$\begin{cases} \dfrac{\partial u}{\partial x}(x,y) = -e^{-\sqrt{n}} \sin(ny) \times n\, ch(nx) \\ \dfrac{\partial^2 u}{\partial x^2}(x,y) = -n^2 e^{-\sqrt{n}} \sin(ny)\, sh(nx) = n^2 u \end{cases}$$

$$\frac{\partial u}{\partial y}(x,y) = -ne^{-\sqrt{n}} sh(nx) \cos(ny)$$
$$\frac{\partial^2 u}{\partial y^2}(x,y) = n^2 e^{-\sqrt{n}} sh(nx) \sin(ny) = -n^2 u$$

$$-\Delta u(x,y) = -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = -n^2 u + n^2 u = 0$$

De plus $u(x,0) = u(x,2\pi) = 0$ $\quad u(0,y) = 0 \quad \dfrac{\partial u}{\partial x}(0,y) = -ne^{-\sqrt{n}} \sin(ny)$

b) On a $|-ne^{-\sqrt{n}} \sin(ny)| \leq ne^{-\sqrt{n}} \xrightarrow[n \to +\infty]{} 0$ CVU (en y) vers 0

or, on vérifie que $u \equiv 0$ sol de
$$\begin{cases} -\Delta u = 0 \\ u(x,0) = u(x,2\pi) = 0 \\ u(0,y) = 0, \ \dfrac{\partial u}{\partial x}(0,y) = 0 \end{cases}$$
(et c'est la seule !)

Pour autant, $-e^{-\sqrt{n}} sh(nx) \sin(ny)$ ne CVU pas vers 0, $n \to +\infty$:
notion de pb mal posé